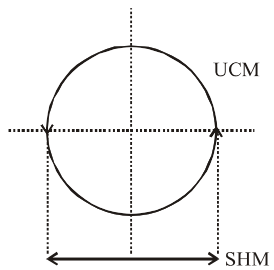
Solutions

2. A polyatomic molecule has many natural frequencies of oscillation. Its vibration is the superposition of individual SHMs.

3. It will be a SHM as projection of uniform circular motion on axis in SHM.



4. $T=\sqrt{\dfrac{\ell}{g}}$, $L_A = L_C$ so $T_A = T_C \Rightarrow n_A = n_C$

So C will be in resonance, hence C will vibrate with maximum amplitude.

5. Term $-kx$ represents restoring force but $-b\bar{v}$ represents damping force.

6. $n_{disp} = n_{velocity} = a_{accn} = n$

but $n_{PE} = 2n$

so $T_{PE} = \dfrac{T}{2} = \dfrac{2}{2} = 1$ sec.

7. $\because$ $x = a\cos(\alpha t)^2$

यह एक कोज्या फलन है जो –a से +a तक परिवर्तित होगा।

It is a cosine function which varies between –a and +a so it is oscillatory. Now for periodic motion.

$x(t + T) = x(t)$ but

$x(t + T) = a\cos(a(t + T))^2$

$= a\cos[\alpha^2t^2 + 2\alpha tT + \alpha^2T^2]$

$\neq x(t)$ so not periodic

8. $TE_{av} = KE_{max.} = PE_{max.} = \dfrac{1}{2}KA^2$

$V_{av} = AW < \cos\omega t > = 0$

but $V_{rms} = \dfrac{V_{max.}}{\sqrt{2}}$

9. $a\sin\omega t + b\cos\omega t = \sqrt{a^2+b^2}\sin(\omega t+\theta)$

Where $\theta = \tan^{-1}\left(\dfrac{b}{a}\right)$ so it is SHM

with amplitude $\sqrt{a^2+b^2}$

10. At $x = \pm A$, $v = 0$ and $a = a_{max}$

11. Velocity is ahead by $\pi/2$ phase angle from displacement

12. $E = \dfrac{1}{2}KA^2e^{-bt/m}$ (exponentially)

13. $b = 0$, $A = \infty$

14. $x = \sin^2\omega t$

$x = \dfrac{1-\cos(2\omega t)}{2}$

$$\boxed{x = \underbrace{\frac{1}{2}}_{\text{Mean position}} - \underbrace{\frac{1}{2}}_{\text{Amplitude}}\cos(2\omega t)}$$

and Timeperiod $= \dfrac{2\pi}{2\omega} = \dfrac{\pi}{\omega}$

15. at t = 2sec. and t = 6 sec oscillater has same consecutime extremes means phased difference is zero.

16. Motion of arrow released from a bow is not periodic (Not repeative motion)

17. In damped oscillations

$$\boxed{x = A_0e^{-\frac{b}{2m}t}\sin(\omega t+\phi)}$$

so graph between disp. time is exponentially decreasing.

18. $F = F_0\cos(\omega_d t)$

19. In forced oscillations -

amplitude $A' = \dfrac{F_0}{\sqrt{m^2(\omega_d^2-\omega^2)^2+\omega_d^2b^2}}$

If $\omega_d = \omega$ than

Amplitude be comes maximum $A_{max} = \dfrac{F_0}{\omega_d b}$

If damping is minimum then peak amplitude maximum.

20. Natural frequency of the body depends on shape and size of body and it material.

21. No. of oscillations per unit time is different means frequencies of both curve are not same.

23. For SHM inertia and elasticity are essential propertes here total energy of system remains conserved.

24. $x = a\cos\omega t$

$y = a\sin\omega t$

$x^2 + y^2 = a^2\cos^2\omega t + a^2\sin^2\omega t = a^2$

$\boxed{x^2+y^2=a^2}$ circle